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SUBJECT CODE NO: - RNP-8046
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Mechanical) (NEP) Sem - IV
Examination May/June-2026
Fluid Mechanics and Fluid Machines

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION – A**Q. 1 Solve any Ten****20**

- 1) Define
 - i. Dynamic viscosity
 - ii. Kinematic viscosity.
- 2) Define
 - i. Surface Tension
 - ii. Capillarity
- 3) Define continuity equation
- 4) Streamlines do not _____ each other.
- 5) What is dimensional homogeneity?
- 6) What is Euler's equation?
- 7) What is venturimeter.
- 8) Define hydraulic gradient line (HGL).
- 9) Define Gross head and Net head
- 10) Define Speed ratio and Jet ratio.
- 11) Priming is required to remove _____ from pump.
- 12) Cavitation occurs due to _____ pressure.

SECTION – B**Q. 2**

- a) Explain variation of viscosity with temperature. **5**
- b) Calculate the dynamic viscosity of an oil, which is used for lubrication between a square plate of size $0.8m \times 0.8m$ & an inclined plane with angle of inclination 30° . The weight of square plate is 300 N & it slides down the inclined plane with a uniform velocity of $0.3m/s$. The thickness of oil film is 1.5 mm. **5**

Q. 3

- a) Classify types of fluid flow with example **5**
- b) Differentiate between streamline, pathline, and streakline. **5**

- Q. 4** a) Derive Euler's equation of motion **5**
 b) An oil of specific gravity 0.8 is flowing through a venturimeter having inlet diameter 20 cm and throat diameter 10 cm. the oil mercury differential manometer shows a reading of 25 cm. Calculate the discharge of oil through the horizontal venturimeter. Take $C_d = 0.98$. **5**
- Q. 5** a) State and explain Buckingham π -theorem. **5**
 b) What is equivalent pipe? Explain with derivation. **5**
- Q. 6** a) Define and explain turbine efficiencies. **5**
 b) Shaft power = 11,772 kW; Head = 380 metres; Speed = 750 r.p.m.; Overall efficiency = 86%; Jet diameter is not to exceed one-sixth of the wheel diameter. Determine: (i) The wheel diameter, (iii) Diameter of the jet. (ii) The number of jets required, and Take $K = 0.985$ and $K = 0.45$ **5**
- Q. 7** a) Describe the working of a centrifugal pump with diagram **5**
 b) The internal and external diameters of the impeller of a centrifugal pump are 100 mm and 400 mm respectively. The pump is running at 1200 r.p.m. The vane angles of the impeller let and outlet are 20° and 30° respectively. The water enters the impeller radially and velocity of flow is constant. Determine the work done by the impeller per unit weight of water. **5**